Web Appendix to "Measuring Capital-Labor Substitution: The Importance of Method Choices and Publication Bias"*

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Abstract

This set of appendices (B–F) provides details on nonlinear tests of publication bias, summary statistics, implementation of BMA, robustness checks, and a list of studies included in the meta-analysis.

Appendix B Furukawa's Method for Addressing Selective Reporting

Furukawa (2019) proposes the so-called stem-based correction method, which relies on the most precise studies, corresponding to the stem of the funnel plot. The method is nonparametric, fully data-dependent and requires weaker assumptions for the underlying distribution of true effects and the publication selection process than other methods. Publication selection can be a function of the size of the estimates, their significance, or both at the same time, as imprecise null results are less likely to be published. By focusing on the n most precise estimates, Furukawa (2019) is able to account for various publication selection processes. The method extends the approach by Stanley *et al.* (2010), who suggest using 10% of the most precise estimates. Instead of selecting an arbitrary number of the most precise estimates, Furukawa (2019) suggests a formal method to calculate the optimal number n of the most precise studies to include by minimizing the mean squared error:

$$\min_n MSE(n) = Bias^2(n) + Var(n). \tag{1}$$

With more studies used, the squared bias term increases as less precise studies suffer from more bias, but the variance term decreases as more information increases efficiency. An empirical analog of the bias term is estimated nonparametrically using two algorithms. The inner algorithm computes the bias-corrected mean given an assumed value of squared precision, and the outer algorithm computes the implied variance and ensures that it is consistent with its

*The manuscript is available at `meta-analysis.cz/sigma`.

assumed value. The inner algorithm ranks and indexes studies in an ascending order according to their standard error, se, and for each $n = 2, ..., N$ calculates the relevant bias squared and variance, given the assumed value of se_0:

$$\tilde{Bias}^2(n) = \frac{\sum_{i=2}^{n} \sum_{j \neq i}^{n} w_i w_j \beta_i \beta_j}{\sum_{i=2}^{n} \sum_{j \neq i}^{n} w_i w_j} - 2\beta_1 \frac{\sum_{i=2}^{n} w_i \beta_i}{\sum_{i=2}^{n} w_i}, \tag{2}$$

$$Var(n) = \sum_{i=1}^{n} w_i, \tag{3}$$

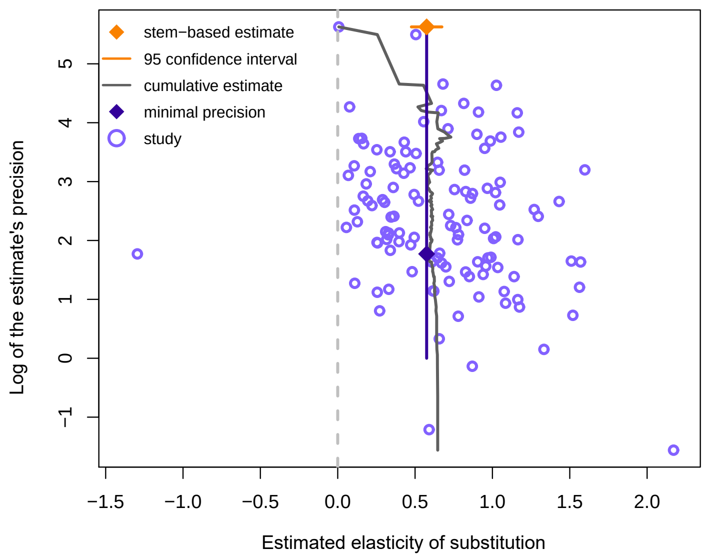
where $w_i = \frac{1}{se_i^2 + se_0^2}$. The optimal number of included studies is given by Equation 1. The stem-based corrected estimate follows:

$$\hat{b}_{stem} = \frac{\sum_{i=1}^{n_{stem}} w_i \beta_i}{\sum_{i=1}^{n_{stem}} w_i}. \tag{4}$$

The outer algorithm then searches over se_0^2 so that the implied variance is consistent.

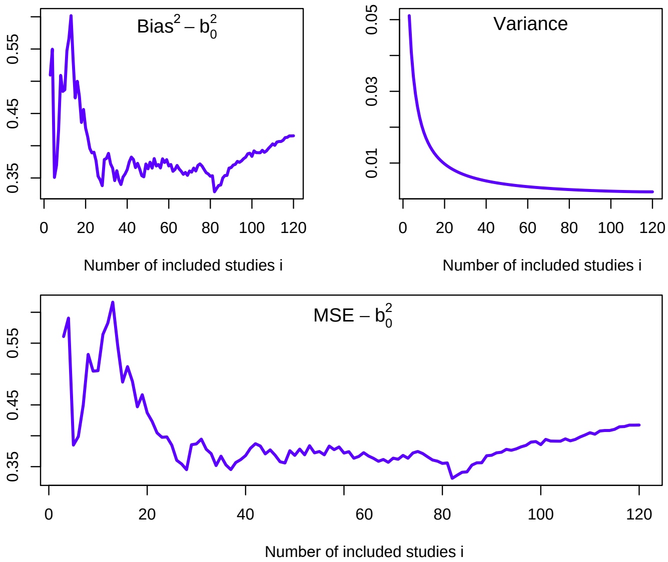
The stem-based method applied to the elasticity of substitution yields the following results: the mean underlying elasticity corrected for publication bias is 0.57 with a standard error of 0.05. Overall, 77% of the total information in the data is utilized, and the 83 most precise studies (out of 121) are included. Because the stem-based method uses study-level estimates (as preferred by Furukawa), we select median values from each study. Figure B1 visualizes the stem-based bias correction method. Figure B2 visualizes the bias-variance trade-off in order to minimize the mean squared error. When all estimates instead of median estimates are used, the mean corrected elasticity is similar, 0.55, but the standard error increases to 0.21.

Figure B1: A graphical illustration of Furukawa's technique



Note: The orange (lighter in grayscale) diamond at the top corresponds to the stem-based estimate of the mean elasticity corrected for publication bias, with the orange line indicating the corresponding 95% confidence interval. The gray (lighter in grayscale) line denotes the estimate under various $n_{stem} \in 1, ..., N$. The blue (darker in grayscale) diamond indicates the minimum precision level that defines the "stem" of the funnel.

Figure B2: The trade-off between bias and variance



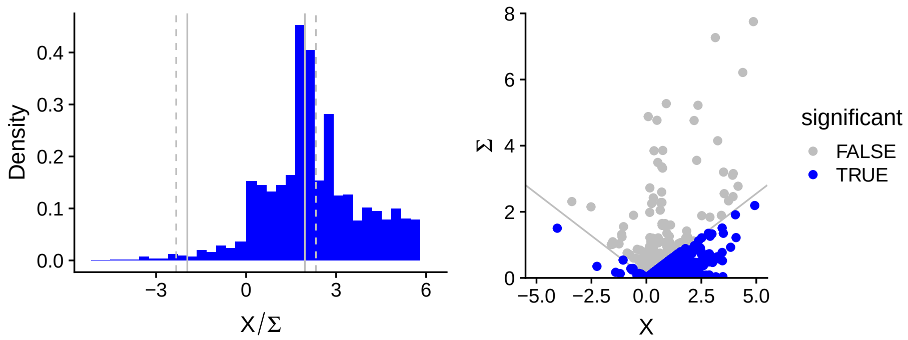
Note: The mean squared error (MSE) is the criterion for choosing the n_{stem}, the optimal number of studies to include in the stem-based estimator. The relevant components of MSE—bias and variance—are plotted.

Appendix C Andrews and Kasy's Method for Addressing Selective Reporting

Andrews & Kasy (2019) introduce two approaches for the identification of publication selection: the first one based on data from replication studies and the second one tailored for meta-analysis. They show that the meta-analysis approach delivers results similar to the approach based on replications. In the absence of publication bias, the distribution of the estimates from imprecise studies can be written as the distribution for precise studies plus noise; deviations from this form identify conditional publication probabilities. Andrews & Kasy (2019) identify publication probability similarly to Hedges (1992) using maximum likelihood: conditional publication probability, $p(\cdot)$, is a step function with jumps at conventional critical values of the p-value.

When applied to our data, the method by Andrews & Kasy (2019) yields the following results. The bias-corrected estimate is 0.43 with a standard error of 0.017. We impose a cutoff at zero, that is, we compare the publication probability of negative vs. positive estimates regardless of their significance. (Allowing for other jumps in publication probability would yield even smaller estimates of the mean elasticity corrected for publication bias.) Our results also suggest that positive estimates are six times more likely to be selected for publication than negative estimates (Table C1). In the case of the elasticity of substitution, publication selection based on statistical significance is apparently less pronounced than selection based on the sign of the estimate, as suggested by the right panel of Figure C1.

Figure C1: A graphical illustration of Andrews and Kasy's (2019) estimator



Note: The solid gray lines mark t-statistic equal to 1.96 in absolute value; the dashed gray line marks t-statistics equal to 2.33 in absolute value. We observe a jump at t-statistic equal to zero and then also jumps at conventional significance levels. The right-hand figure plots estimates X and their standard errors Σ; the gray line marks 1.96 in absolute value. Even though we observe discontinuity at the t-statistic corresponding to the 5% significance level, the right panels shows publication selection based on significance is not absolute, as some insignificant estimates (gray points) are reported.

Table C1: Results of Andrews and Kasy's (2019) estimator

	$\bar{\theta}$	$\bar{\tau}$	DF	β_p
Estimate	0.430	0.489	12.809	0.158
Standard error	0.017	0.012	0.707	0.019

Notes: $\bar{\theta}$ denotes the bias-corrected mean effect, $\bar{\tau}$ is a scale parameter, DF are degrees of freedom. β_p is a publication probability measured relative to the omitted category, in our case positive estimates. An estimate of 0.158 therefore implies that negative results are 15.8% as likely to be published as positive ones.

Appendix D Description of Variables

Table D1: Definitions and summary statistics of explanatory variables

Variable	Description	Mean	Std. dev.
Data characteristics			
No. of obs.	The logarithm of the number of observations used in the regression.	4.28	1.51
Midpoint	The logarithm of the mean year of the data used minus the earliest mean year in our data.	4.71	0.48
Cross-sec.	= 1 if cross-sectional data are used (reference category: time series).	0.33	0.47
Panel	= 1 if panel data are used (reference category: time series).	0.14	0.35
Quarterly	= 1 if the data frequency is quarterly (reference category: annual).	0.11	0.31
Industry data	= 1 if variation at the industry-/sector-level is exploited in input data (reference category: cross-country-/state-level variation).	0.43	0.50
Firm data	= 1 if variation at the firm-level is exploited in input data (reference category: cross-country-/state-level variation).	0.12	0.32
Country: US	= 1 if the estimate is for the US.	0.58	0.49
Country: Eur	= 1 if the estimate is for a developed European country.	0.17	0.37
Developing	= 2 if the estimate is for a developing country; = 1 if the estimate is a common estimate for a collection of developed and developing countries (reference category: developed countries).	0.22	0.54
Database: OECD	= 1 if the data come from the OECD database.	0.07	0.25
Database: KLEM	= 1 if the data come from the Jorgenson KLEM dataset.	0.15	0.36
Database: ASMCM	= 1 if the data come from the Annual Survey of Manufacturers and/or Census of Manufacturers.	0.14	0.35
Disaggregated σ	= 1 if the elasticity is estimated on a disaggregated level (industry-specific elasticity).	0.52	0.50
Specification			
System PF-FOC	= 1 if the elasticity is estimated within a system of CES with FOC(s) or with cost share functions.	0.06	0.23
System FOCs	= 1 if the elasticity is estimated within a system of FOCs.	0.05	0.23
Nonlinear	= 1 if the elasticity is estimated within the CES directly via nonlinear methods.	0.04	0.20
Linear approx.	= 1 if the elasticity is estimated via Taylor series expansion (Kmenta approach or translog approach).	0.07	0.26
FOC_L_w	= 1 if the elasticity is estimated within the FOC for labor based on the wage rate (reference category: FOC for capital based on the rental rate of capital).	0.33	0.47
FOC_KL_rw	= 1 if the elasticity is estimated within the FOC of K/L based on w/r (reference category: FOC for capital based on the rental rate of capital).	0.18	0.39
FOC_K_share	= 1 if the elasticity is estimated within the FOC for capital based on the capital share (reference category: FOC for capital based on the rental rate of capital).	0.03	0.16
FOC_L_share	= 1 if the elasticity is estimated within the FOC for labor based on the labor share (reference category: FOC for capital based on the rental rate of capital).	0.04	0.19
User cost elast.	= 1 if the user cost of capital elasticity is estimated.	0.17	0.38
Cross-equation rest.	= 1 if cross-equation restrictions are employed when using system estimation.	0.08	0.28
Normalized	= 1 if normalization is applied to the CES.	0.05	0.22

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Table D1: Definitions and summary statistics of explanatory variables (continued)

Variable	Description	Mean	Std. dev.
Two-level PF	= 1 if a two-level CES function is estimated (due to more than two factors of production).	0.03	0.18
Partial σ	= 1 if some form of partial elasticity is used (Allen-Uzawa, Hicks-Allen, Morishima).	0.06	0.24
Econometric approach			
Dynamic est.	= 1 if dynamic methods are used for estimation (VAR, a distributed lag model or error correction model; reference category: OLS).	0.24	0.42
SUR	= 1 if a system of seemingly unrelated regressions is used (Zellner's estimation; reference category: OLS).	0.11	0.31
Differenced	= 1 if the coefficient is taken from a regression in first differences or log differences.	0.23	0.42
Time FE	= 1 if time-fixed effects are used for estimation.	0.06	0.24
Unit FE	= 1 if unit-fixed effects are used for estimation.	0.04	0.20
Identification	= 1 if instrumental variables are used for identification.	0.13	0.34
Short-run σ	= 1 if the coefficient is taken from an explicit short-run specification (reference category: explicit long-run specification—cointegration, low-pass filter, interval-difference model).	0.05	0.22
Long-run σ unadj.	= 1 if the coefficient is meant to be long-run but the specification is not adjusted accordingly (reference category: explicit long-run specification).	0.68	0.47
Production function components			
Other inputs in PF	= 1 if the production function includes other inputs such as energy, materials, and human capital.	0.13	0.34
LATC	= 1 if the production function includes labor-augmenting technological change, i.e. Harrod-neutral technological change (reference category: Hicks-neutral technological change).	0.29	0.63
CATC	= 1 if the production function includes capital-augmenting technological change, i.e. Solow-neutral technological change (reference category: Hicks-neutral technological change).	0.26	0.57
Skilled L	= 1 if the production function distinguishes between skilled and unskilled labor.	0.02	0.13
Constant TC growth	= 1 if the technological change is modeled with constant growth rates (reference category: no growth of technology).	0.30	0.46
Other TC growth	= 1 if the technological change is modeled with nonconstant growth rates, e.g., logarithmic, linear (reference category: no growth of technology).	0.10	0.31
No CRS	= 1 if the authors assume nonconstant returns to scale.	0.09	0.36
No full comp.	= 1 if the authors do not assume factor markets to be perfectly competitive.	0.04	0.19
Net σ	= 1 if net elasticity is estimated (reference category: gross elasticity).	0.02	0.16
External info			
Top journal	= 1 if the study is published in a top five journal in economics.	0.31	0.46
Pub. year	The logarithm of the year when the first draft of the study appeared in Google Scholar minus the year when the first study on elasticity of substitution was written.	3.25	0.88
Impact	The recursive discounted RePEc impact factor of the outlet.	0.96	1.07
Citations	The logarithm of the number of per-year citations of the study since its first appearance on Google Scholar.	1.47	0.96
Preferred est.	= 1 if the estimate is preferred by authors or is explicitly considered to be better; -1 if it is considered inferior.	-0.04	0.47
Byproduct	= 1 if estimation of the elasticity is not the central focus of the paper but only a byproduct; = 0 if it is the central focus; = 0.5 if it is one of multiple main aims.	0.20	0.31

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Table D1: Definitions and summary statistics of explanatory variables (continued)

Variable	Description	Mean	Std. dev.
Measurement of variables			
y: index	= 1 if the input data for total output is in an index form.	0.03	0.18
y: other	= 1 if the input data for total output is measured differently than in gross domestic product or total value added (reference category: GDP, value added).	0.07	0.26
Labor-related			
Quality adj.	= 1 if the input data for labor incomes data are quality-adjusted.	0.22	0.41
Self empl.	= 1 if the input data for labor incomes data are adjusted for the income of self-employed people.	0.18	0.39
w: nominal	= 1 if the input data for the wage rate are nominal (reference category: the wage rate is in real terms).	0.09	0.29
w: direct	= 1 if the input data for the wage rate are measured directly (the wage rate calculated as total wages divided by the total number of employees).	0.14	0.36
L: hours	= 1 if the input data for the labor are measured in hours.	0.25	0.44
L: years	= 1 if the input data for the labor are measured in years.	0.07	0.25
L: FTE workers	= 1 if the input data for labor are measured by the full-time equivalent number of workers.	0.07	0.25
L: force	= 1 if number of workers labor is measured as the total number of people in the labor force.	0.04	0.20
Capital-related			
Capacity adj.	= 1 if the authors control for the capacity utilization in the regression.	0.09	0.28
r: quasi	= 1 if the input data for the rental rate of capital are measured as the quasi-rent, i.e., total output minus total wages divided by total capital stock (reference category: it is measured as the user cost of capital, Hall-Jorgenson formula).	0.24	0.43
r: nominal	= 1 if the input data for the rental rate of capital are expressed in nominal terms.	0.01	0.09
K: IT	= 1 if IT capital is used only.	0.02	0.13
K: equipment	= 1 if the measure of equipment capital is used only.	0.07	0.26
K: structures	= 1 if the measure of structures, land or plant is used only.	0.04	0.17
K: residential	= 1 if the measure of capital includes residential capital stock.	0.07	0.25
K: services	= 1 if capital is measured as service flow.	0.13	0.33
K: perpetual	= 1 if the input data for capital is measured via perpetual inventory method.	0.36	0.48
K: index	= 1 if the input data for capital are expressed in an index form.	0.17	0.37
Industry-related			
Primary ind.	= 1 if the elasticity is estimated for the primary sector.	0.02	0.14
Secondary ind.	= 1 if the elasticity is estimated for the secondary sector.	0.62	0.49
Tertiary ind.	= 1 if the elasticity is estimated for the tertiary sector.	0.03	0.18
Materials	= 1 if the elasticity is estimated for the 2-digit industry in the category "Materials" of the GICS industry classification.	0.25	0.43
Industrials	= 1 if the elasticity is estimated for the 2-digit industry in the category "Industrials" of the GICS industry classification.	0.09	0.29
Consumer	= 1 if the elasticity is estimated for the 2-digit industry in the category "Consumer goods" of the GICS industry classification.	0.14	0.34

Note: Collected from published studies estimating the elasticity of substitution between capital and labor. When dummy variables form groups, we mention the reference category.

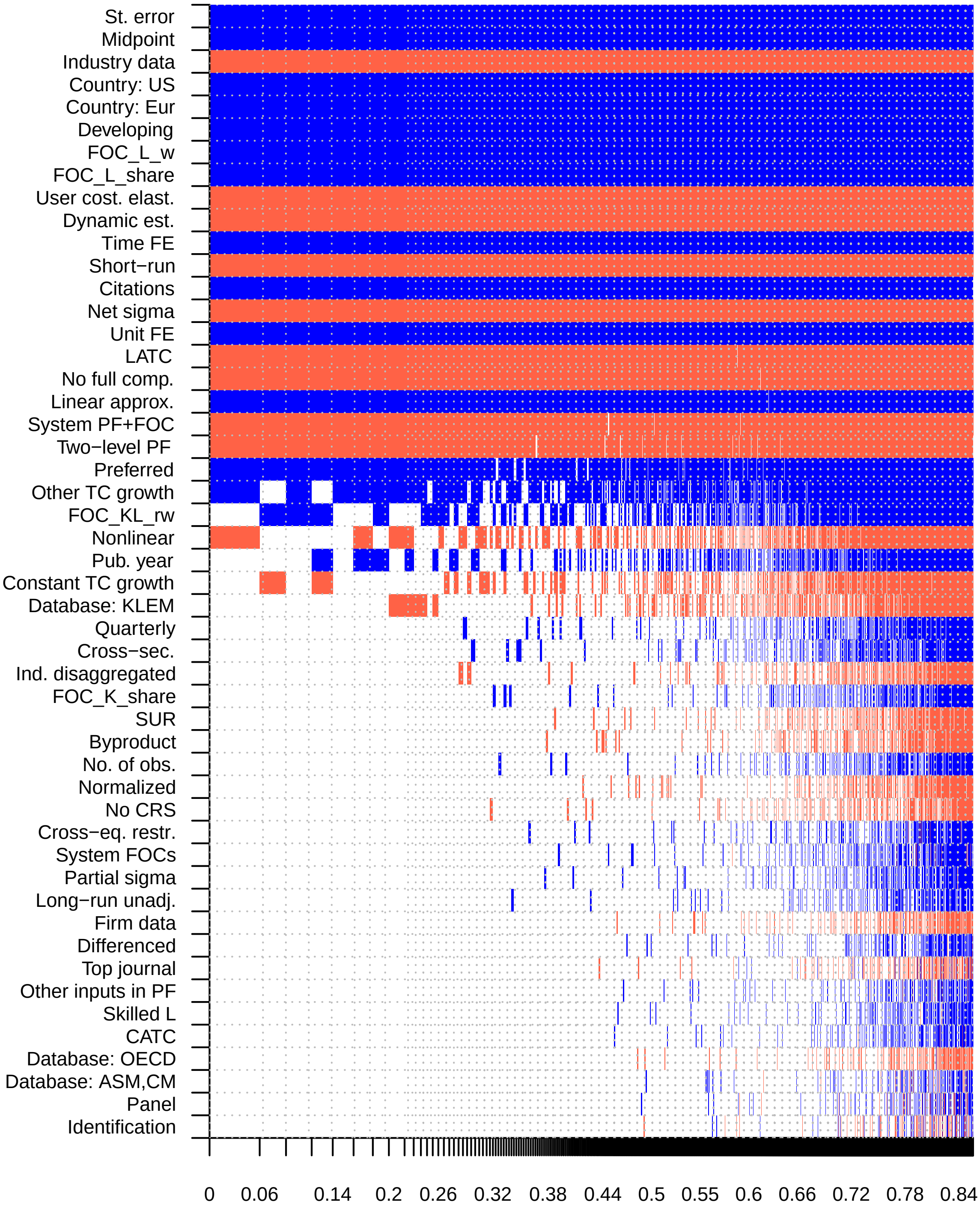
Appendix E Robustness Checks

Table E1: Results of frequentist model averaging

	Coef.	Std. er.	p-value
Standard error	0.557	0.042	0.000
Data characteristics			
No. of obs.	0.011	0.012	0.326
Midpoint	0.103	0.022	0.000
Cross-sec.	0.069	0.029	0.016
Panel	0.193	0.042	0.000
Quarterly	0.135	0.042	0.001
Firm data	-0.160	0.040	0.000
Industry data	-0.198	0.026	0.000
Country: US	0.121	0.031	0.000
Country: Eur	0.180	0.030	0.000
Developing	0.019	0.019	0.333
Database: ASM,CM	-0.031	0.037	0.402
Database: OECD	-0.301	0.044	0.000
Database: KLEM	-0.092	0.046	0.047
Disaggregated σ	0.043	0.024	0.077
Specification			
System PF+FOC	-0.111	0.059	0.061
System FOCs	-0.057	0.050	0.258
Nonlinear	-0.016	0.061	0.796
Linear approx.	0.268	0.050	0.000
FOC_L_w	0.324	0.032	0.000
FOC_KL_rw	0.007	0.032	0.832
FOC_K_share	0.226	0.063	0.000
FOC_L_share	0.251	0.048	0.000
Cross-eq. restr.	0.071	0.048	0.140
Normalized	-0.248	0.051	0.000
Two-level PF	-0.023	0.070	0.743
Partial sigma	0.130	0.055	0.018
User cost. elast.	-0.373	0.042	0.000
Econometric approach			
Dynamic est.	-0.005	0.029	0.854
SUR	-0.105	0.032	0.001
Identification	0.046	0.026	0.077
Differenced	-0.096	0.027	0.000
Time FE	-0.009	0.040	0.830
Unit FE	0.067	0.043	0.116
Short-run	-0.410	0.040	0.000
Long-run unadj.	-0.011	0.026	0.681
Production function components			
Other inputs in PF	-0.137	0.044	0.002
CATC	-0.003	0.026	0.904
LATC	-0.041	0.024	0.088
Skilled L	0.076	0.059	0.199
Constant TC growth	-0.032	0.025	0.191
Other TC growth	0.108	0.035	0.002
No CRS	-0.003	0.022	0.905
No full comp.	-0.022	0.042	0.598
Net sigma	-0.320	0.056	0.000
Publication characteristics			
Top journal	-0.085	0.025	0.001
Pub. year	0.032	0.015	0.038
Citations	0.037	0.011	0.001
Preferred	0.027	0.016	0.093
Byproduct	-0.130	0.032	0.000
(Intercept)	-0.123	0.130	0.342
Observations	3,186		

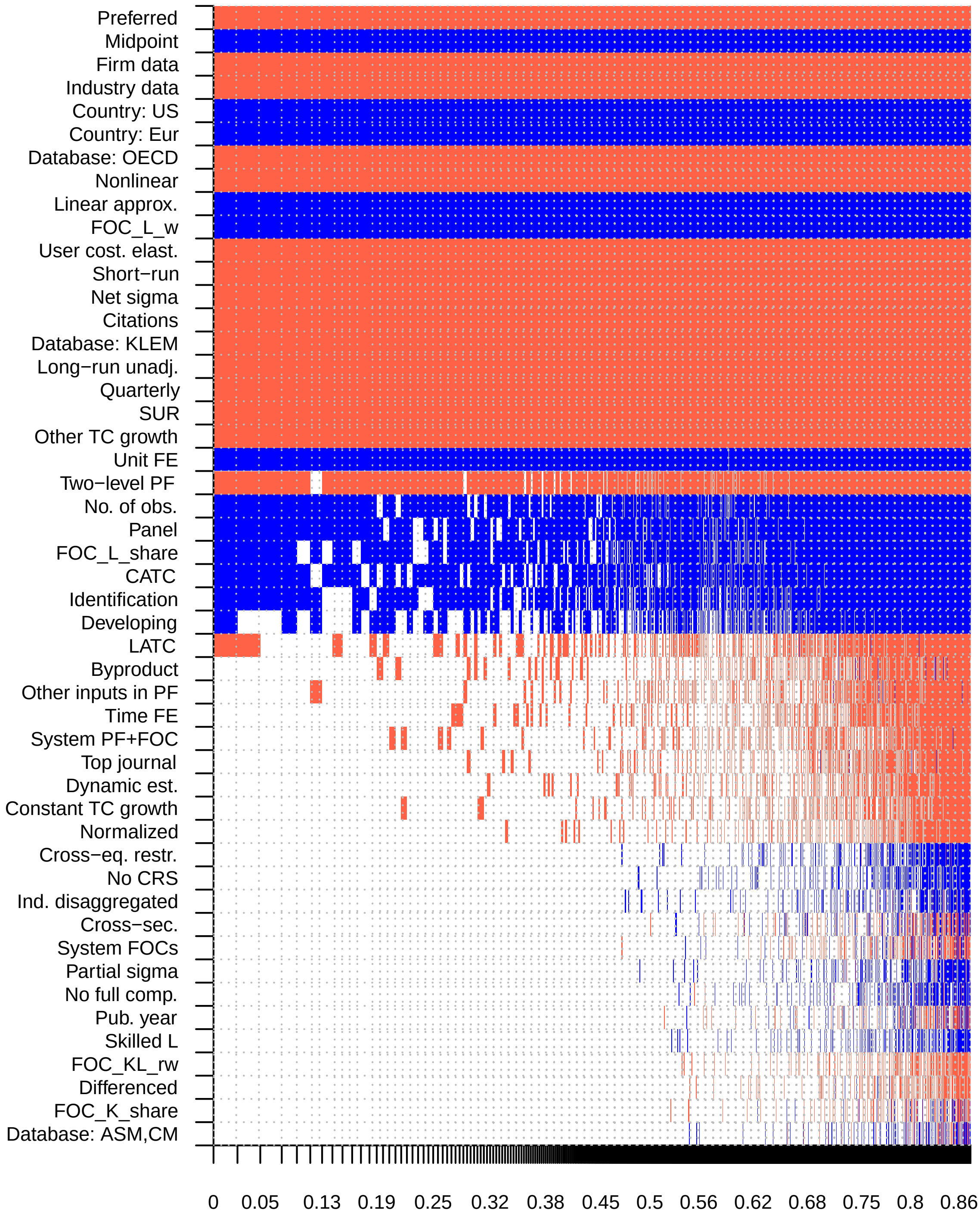
Notes: Our frequentist model averaging (FMA) exercise employs Mallow's weights (Hansen, 2007) and the orthogonalization of the covariate space suggested by Amini & Parmeter (2012). Dark gray color denotes variables that are deemed important also in the BMA exercise. Light gray color denote variables that are deemed important in the FMA but not BMA exercise.

Figure E1: Model inclusion in Bayesian model averaging, weighted by the inverse of the number of estimates per study



Notes: The response variable is the estimate of the elasticity of substitution. Columns denote individual models; variables are sorted by posterior inclusion probability in descending order. The horizontal axis denotes cumulative posterior model probabilities; only the 5,000 best models are shown. Blue color (darker in grayscale) = the variable is included and the estimated sign is positive. Red color (lighter in grayscale) = the variable is included and the estimated sign is negative. No color = the variable is not included in the model.

Figure E2: Model inclusion in Bayesian model averaging, weighted by the inverse of the standard error



Notes: The response variable is the estimate of the elasticity of substitution. Columns denote individual models; variables are sorted by posterior inclusion probability in descending order. The horizontal axis denotes cumulative posterior model probabilities; only the 5,000 best models are shown. Blue color (darker in grayscale) = the variable is included and the estimated sign is positive. Red color (lighter in grayscale) = the variable is included and the estimated sign is negative. No color = the variable is not included in the model.

Subsamples with measurement variables

As a complementary exercise to our baseline specification, we also run BMA analyses for subsamples of data in order to control for variables that are relevant only for a given subsample. We call these variables measurement variables. We need to create subsamples of the main dataset, because the variables relevant for the FOC for labor are not relevant for the FOC for capital, and vice versa. Regarding the estimates that utilize the FOC for labor, we include additional variables on how labor and the wage rate are measured. Regarding the estimates that utilize the FOC for capital, we include variables on how capital and the rental rate of capital are measured. Regarding industry-level estimates, we include the sector for which the elasticity was estimated, that is, primary, secondary and tertiary sectors; and, within the secondary sector, groups for industrial goods production, material goods production, and consumer goods production.

Concerning the measurement of labor, our reference category is measurement via the number of workers. We include a dummy equal to one if labor is measured using the number of hours worked. We also include a dummy variable that equals one if labor income is adjusted for self-employed labor income. As for the wage rate, we include dummy variables for the case when the rate is measured directly (in contrast to the situation when the wage rate is measured as the total amount paid to employees divided by the labor variable) and when the wage rate is used in nominal terms. In addition, we examine the effect of adjusting for changes in skill over time, for example, adjusting for the share of white- versus blue-collar workers.

Concerning the measurement of capital, our reference category is unspecified capital. We include dummies for specific measurements, including measurement as service flow, measurement via the perpetual inventory method, and capital stock in an index form. We code for special categories of capital stock: equipment, structures, IT, and residential capital stock. We include a separate dummy equal to one if the study controls for capacity utilization, either by adjusting the measurement variables or by adding it as a control. Underutilized capital would bias the results since it biases the effect of input on output (Brown, 1966); nevertheless, only a small portion of studies (Brown, 1966; Behrman, 1972; Dissou *et al.*, 2015, among others) explicitly use this approach, for example by including capacity utilization indices.

Regarding the rental rate of capital, the baseline category comprises the user cost of capital, or, in other words, the standard Hall-Jorgenson formula (Jorgenson, 1963; Hall & Jorgenson, 1967), which appears in two-thirds of all the estimations. The Hall-Jorgenson formula calculates the user cost of capital as a function of the relative price of capital, rate of return, and depreciation. We include a dummy for the case when the tax rate is an additional variable in the Hall-Jorgenson formula. The second most frequently used measurement is the quasi-rent approach, which calculates the rental rate of capital as a difference between total value added and total wages divided by the capital stock; this approach is used in 17% of the cases, for example in Dhrymes (1965), Ferguson (1965), and Lovell (1973). Further, the rental rate of capital can be measured either in gross terms or in net terms and in real or nominal terms; nevertheless, the variability in nominal user cost is almost zero, and thus we do not include the corresponding variable.

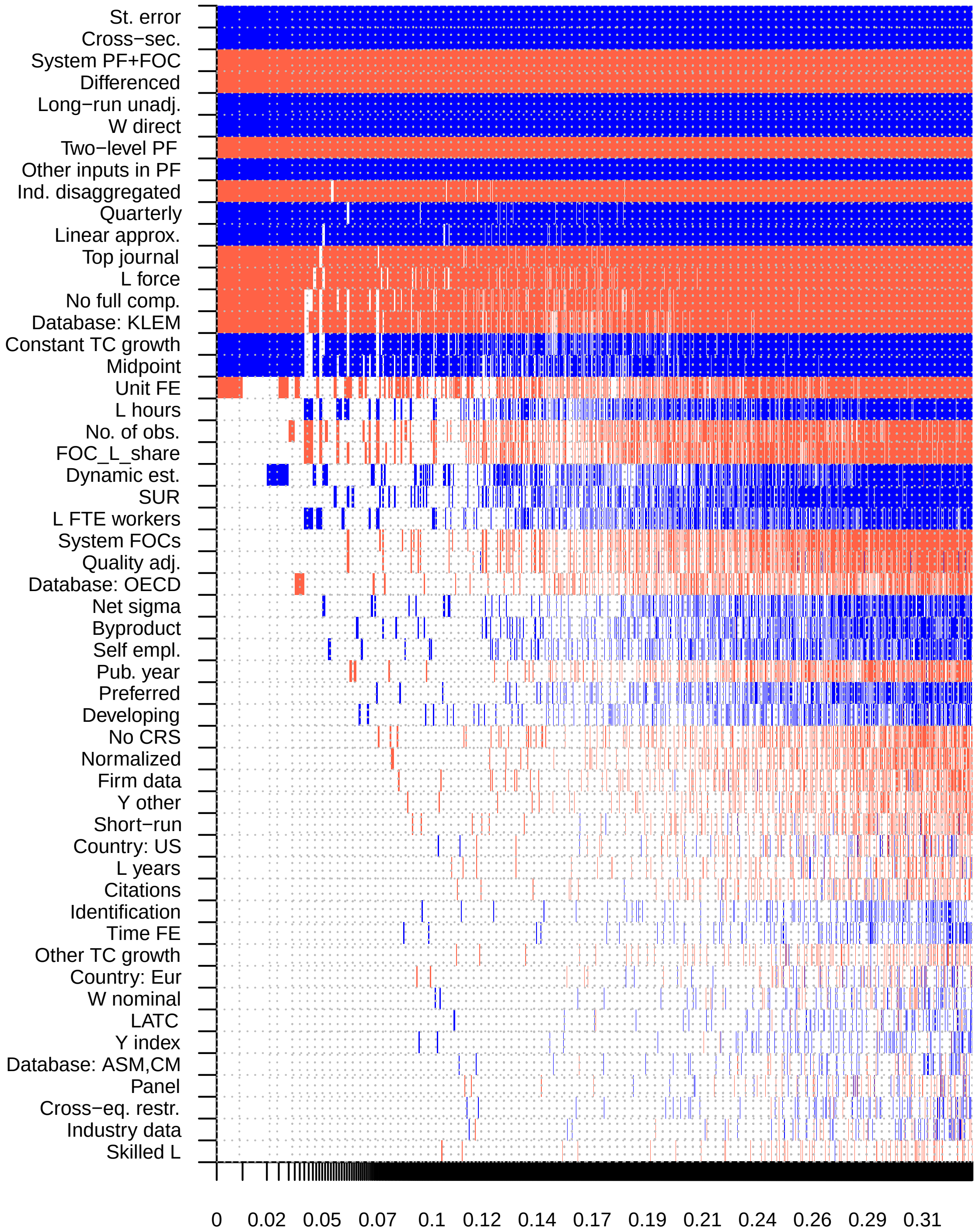
In all subsamples we control for the measurement of output: first, we include a dummy variable that equals one if output is not measured as gross product or in value added terms, but in another way—for example, as the amount of sales. Second, we include a dummy for the case when output is used in an index form.

How does the addition of these variables affect our results? First, we include labor-specific variables, which capture how labor and wage rate are measured, and run BMA on the subsample of data estimating the FOC for labor. The subsample covers less than half of the original dataset; the results are displayed in Figure E3. Only two of the newly included measurement variables are important for the explanation of the heterogeneity in the reported elasticities: direct measurement of the wage rate and measurement of labor as total labor force. The main drivers of heterogeneity remain the same while the total explanatory power of the analysis increases only marginally.

Concerning capital-related variables, we find that the type of capital under examination represents an important driver of the differences in results (Figure E4). IT capital and equipment capital are more substitutable with labor than other types of capital, such as buildings. When capital is measured as service flow, the estimates typically yield a larger elasticity of substitution. It also matters how the rental rate of capital, r, is computed, specifically whether the Hall-Jorgenson formula is used—we find that it yields smaller elasticities than do other approaches. The best-practice estimate derived from both subsamples and conditional on plugging in mean values for measurement variables would again equal 0.3, far from the Cobb-Douglas assumption.

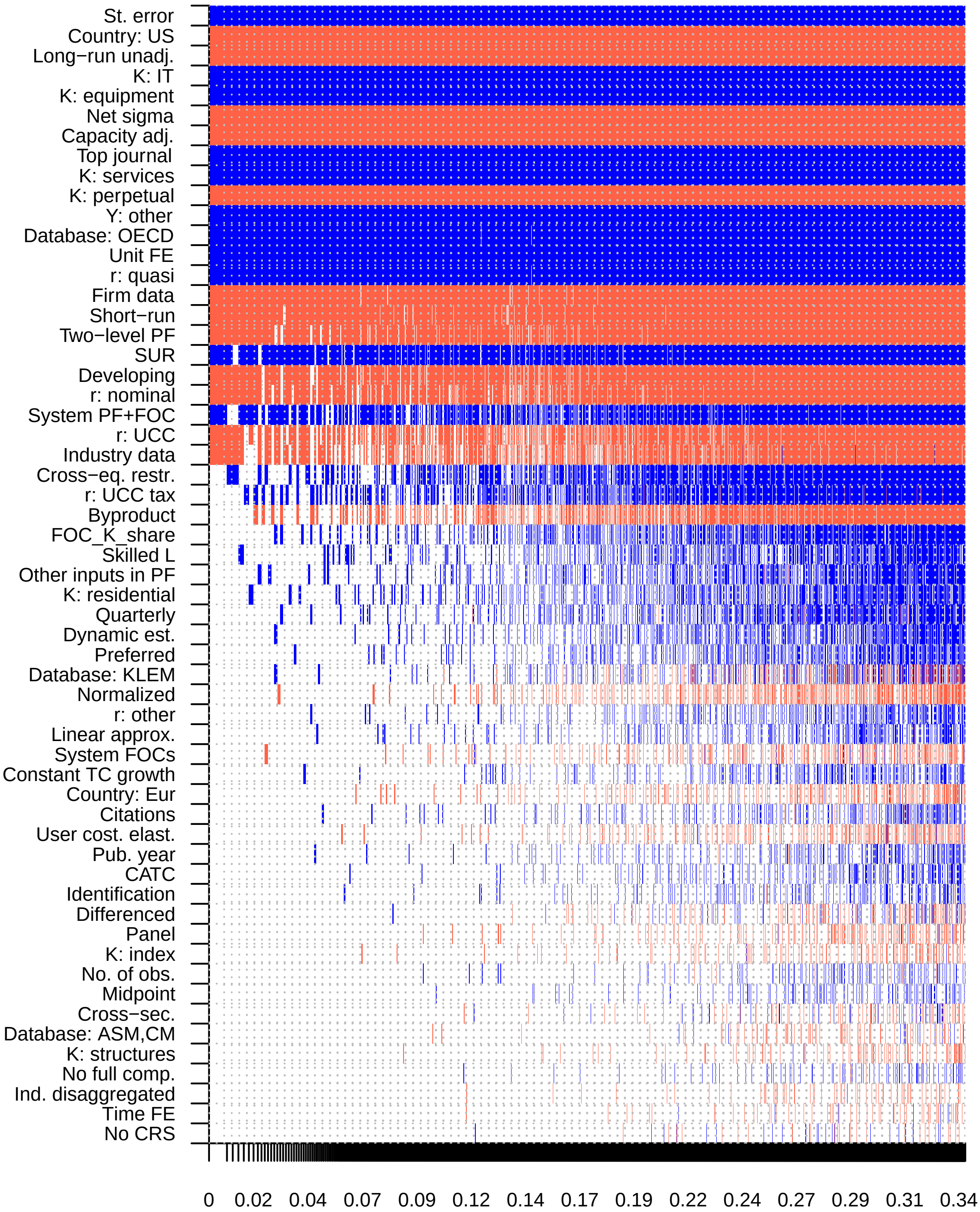
Finally, for the subsample of disaggregated elasticities we run the baseline BMA enriched with industry-relevant variables in Figure E5. We do not find any significant determinants that would suggest that the elasticity of capital-labor substitution differs systematically across sectors or industry groups (production of materials, production of industrial goods, production of consumer goods, and production of services). Given the number of variables in our analysis, it is infeasible to add more industry-specific variables since that would create troubles with collinearity.

Figure E3: Model inclusion in Bayesian model averaging, labor-specific variables



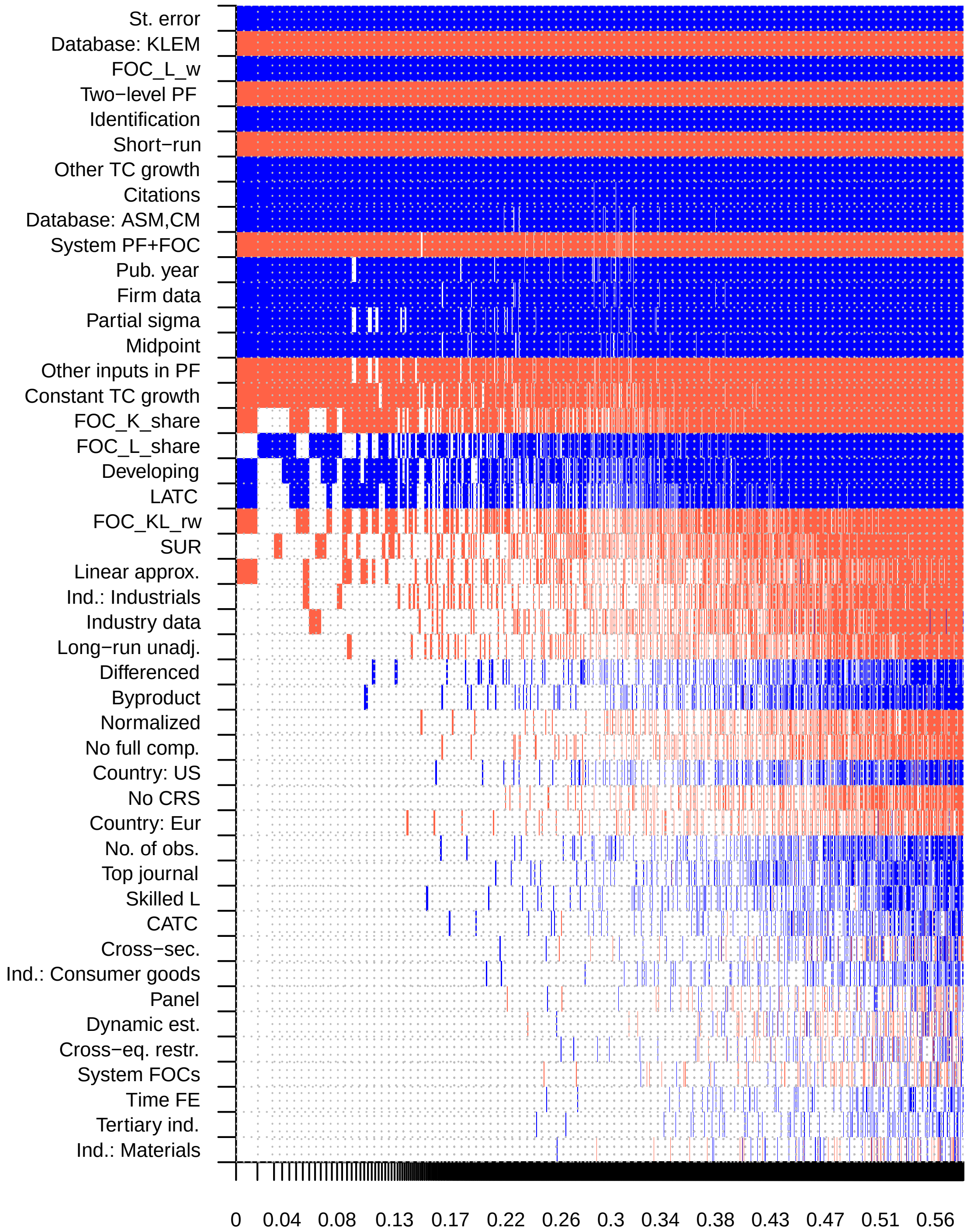
Notes: The response variable is the estimate of the elasticity of substitution. Columns denote individual models; variables are sorted by posterior inclusion probability in descending order. The horizontal axis denotes cumulative posterior model probabilities; only the 5,000 best models are shown. Blue color (darker in grayscale) = the variable is included and the estimated sign is positive. Red color (lighter in grayscale) = the variable is included and the estimated sign is negative. No color = the variable is not included in the model.

Figure E4: Model inclusion in Bayesian model averaging, capital-specific variables



Notes: The response variable is the estimate of the elasticity of substitution. Columns denote individual models; variables are sorted by posterior inclusion probability in descending order. The horizontal axis denotes cumulative posterior model probabilities; only the 5,000 best models are shown. Blue color (darker in grayscale) = the variable is included and the estimated sign is positive. Red color (lighter in grayscale) = the variable is included and the estimated sign is negative. No color = the variable is not included in the model.

Figure E5: Model inclusion in Bayesian model averaging, industry-specific variables



Notes: The response variable is the estimate of the elasticity of substitution. Columns denote individual models; variables are sorted by posterior inclusion probability in descending order. The horizontal axis denotes cumulative posterior model probabilities; only the 5,000 best models are shown. Blue color (darker in grayscale) = the variable is included and the estimated sign is positive. Red color (lighter in grayscale) = the variable is included and the estimated sign is negative. No color = the variable is not included in the model.

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